

Annex 3: Arbitrary Baud Rate Calculation and Configuration

Instruction Manual

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I. Use of the Baud Rate Calculation Tool

In order to easily calculate the baud rate of our USBCAN adapter/CANalyst-II analyzer, you can use the tool software shown in Figure 1. (Placed in the USB_CAN TOOL software installation directory C:\Program Files (x86)\USB_CAN TOOL)

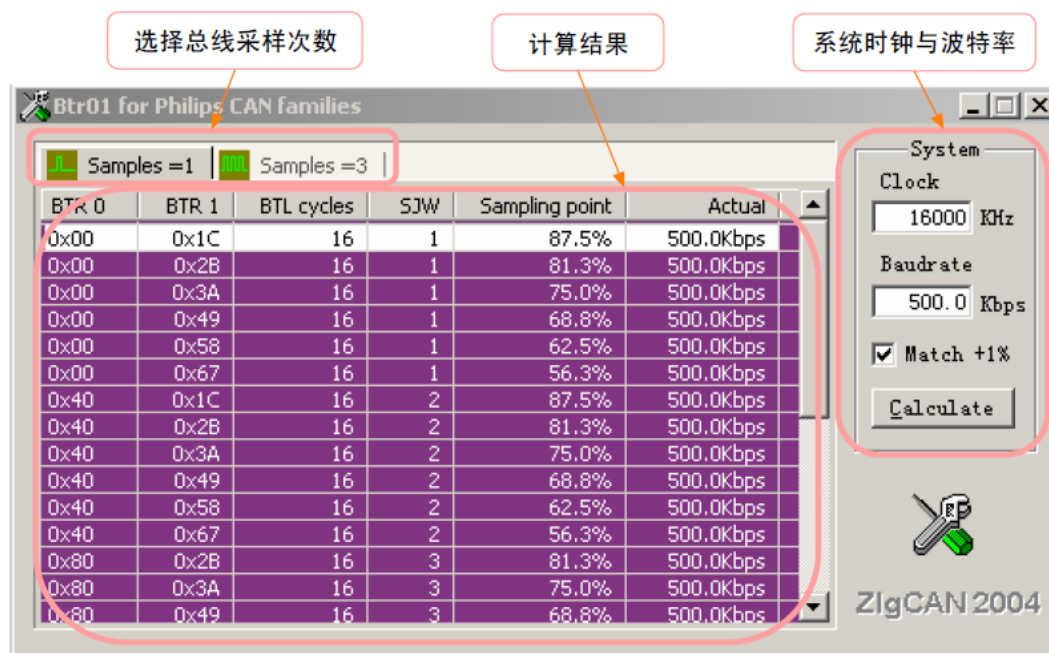


Figure 1 CAN Baud Rate Calculation Software

This tool software is simple and convenient, according to the following steps can easily realize the baud rate calculation.

- Fill in the system clock of CAN controller in the first box (clock) on the right side of the software. For our products, the crystal oscillator is 16Mhz, then you need to fill in "16000";
- In the second box (Baudrate) on the right side of the software, fill in the actual baud rate required by the user. For example, if you need a baud rate of 500Kbps, you need to fill in "500";
- If the calculated baud rate is allowed to have 1% error (according to the CAN protocol 1% error is within the tolerance range), then you can check the checkbox "Match +1%";
- Selects the sampling rate (samples) in the upper left corner of the software. Sample rate can choose to sample once or three times, usually in the case of low

baud rate (less than 40Kbps) to use three samples, which can effectively eliminate bus burrs. At high baud rates, one sample is usually used;

- Once the baud rate requirements above are filled in, click on the Calculate button (Calculate) and the baud rate used by the user will be calculated in the table on the left.
- The result listed in the table consists of the six parameters shown in Figure 2, which are represented as follows.

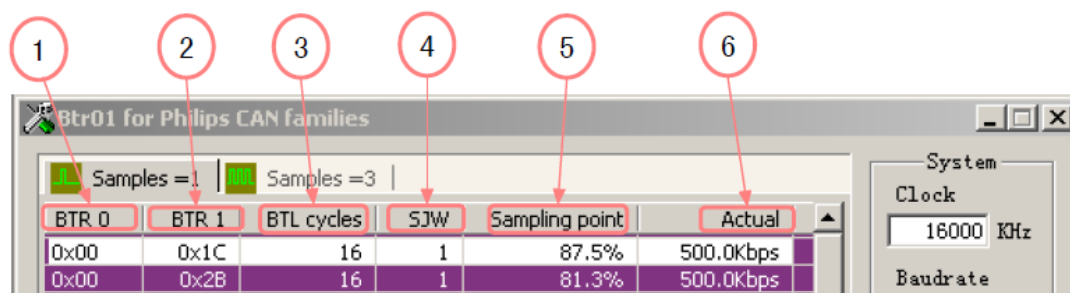


Figure 2 Illustration of table columns

1. BTR0, Bus Timer 0, where the high two bits are the synchronization jump width and the low six bits are baud rate preset;
2. BRT1, Bus Timer 1, where the lower four bits are Phase Buffer Segment 1, every fourth to sixth bit is Phase Buffer Segment 2, and the highest bit is the sample count selection;
3. BTLcycles, the number of unit time shares occupied by the bus timer, bus timer length = unit time * BTLCycle, i.e., there is BTLCycle = (synchronization segment + propagation segment + phase buffer segment 1 + phase buffer segment 2), usually synchronization and propagation segments for 1 time share;
4. SJW, bus resynchronization jumper width, usually its value range is 1~4;
5. sampling point, the position of the bus sampling point, i.e., the position of the sampling point occupying the whole bit time when sampling the bus;
6. Actual, the exact baud rate value calculated according to the set of data.

The values listed in the table all meet the requirements according to the CAN protocol, and the user can choose any set of data as the current baud rate value. However, if there is no data calculated in the table after clicking the Calculate button,

it means that according to the crystal oscillator frequency set by the user and the required baud rate, the result can not be calculated to meet the requirements, in this case, the user can try to change the baud rate to recalculate.

Open USB_CAN TOOL, click "Start Device" in the "Device Operation" menu. In the "Parameter Confirmation" dialog box, select "Self Define" for the baud rate drop-down list. In the "Parameter Confirmation" dialog box, select "Self Define" in the Baud Rate drop-down list. Fill the two bytes (hexadecimal) of BTR0 and BTR1 calculated by the Baud Rate Calculator into the two input boxes of BTR0/1 in the "Parameter Confirmation" dialog box of USB_CAN TOOL software. As shown in the figure, configure the 500k baud rate calculated in the above figure.

Caution, the two channels are completely independent and the baud rate needs to be configured separately by selecting the corresponding channel via the "Select CAN channel number" drop-down list.

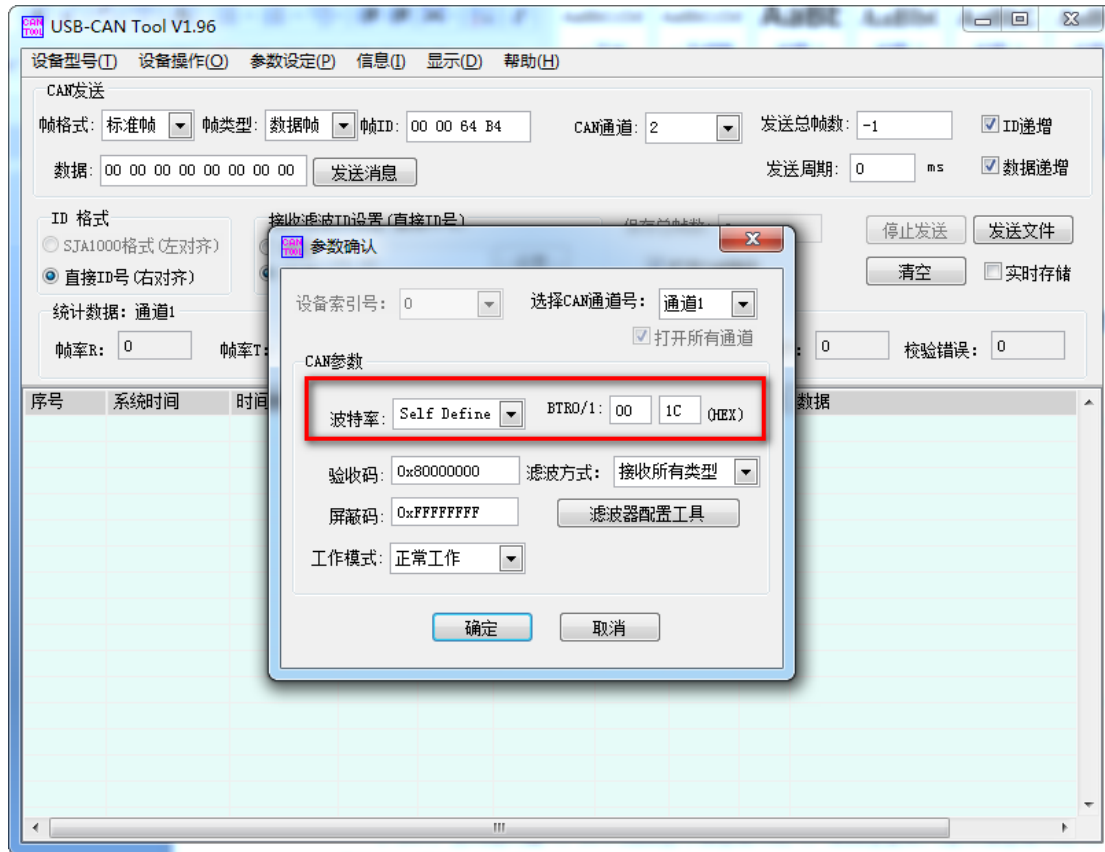


Figure 3 Baud rate configuration

Caution: This software is only suitable for calculating the baud rate of our products, e.g. USBCAN adapter/CANalyst-II. The data calculated by this software may not be applicable to CAN controllers of other companies.

II. CAN Baud Rate Basics

The SJA1000 internal frequency reference source $F_{BASE} = F_{clk}/2$, which is the 2-division frequency of the external crystal frequency F_{clk} . It's worth noting that any application that utilizes an external crystal as a reference source is first shaped by a 2-division frequency.

1. Composition of the Bit Cycle

The baud rate (f_{bit}) is the number of data bits transmitted per unit of time, generally taken as 1s per unit of time. The baud rate is determined by the length of one data bit cycle (T_{bit}) transmitted over the communication line, as shown in the following equation:

$$F_{bit} = 1/T_{bit} \quad (1)$$

According to Philips' stand-alone communication controller, a bit cycle consists of 3 parts: the synchronization segment (t_{SYNC_SEG}), the phase buffer segment 1 (t_{TSEG1}) and the phase buffer segment 2 (t_{TSEG2}).

$$T_{bit} = t_{SYNC_SEG} + t_{TSEG1} + t_{TSEG2} \quad (2)$$

All of these time periods have a common unit of time - the system clock period (TSCL). Specifically for the SJA1000, the TSCL is determined by the value of the Bus Timing Register. The SJA1000 has 2 Bus Timing Registers, Bus Timing Register 0 (BTR0) and Bus Timing Register 1 (BTR1), which have their own different functional definitions and work together to determine the communication baud rate of the bus.

2. Bus Timing Register

Bus Timing Register 0 Defines the values of the baud rate preset BRP (6 bits in total, value interval [1,64]) and the synchronization jump width SJW (2 bits in total, value interval [1,4]). The bit function descriptions are listed in Table 1.

Table 1: Bus Timing Register 0 (BTR0) Bit Function Description

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SJW.1	SJW.0	BRP.5	BRP.4	BRP.3	BRP.2	BRP.1	BRP.0

The system clock period T_{SCL} of the CAN can be determined by the value of BRP as, calculated as follows:

$$T_{SCL} = 2 \times T_{CLK} \times BRP = 2 \times T_{CLK} \times (32 \times BRP.5 + 16 \times BRP.4 + 8 \times BRP.3 + 4 \times BRP.2 + 2 \times BRP.1 + 1 \times BRP.0 + 1) \quad (3)$$

T_{CLK} is the period of the reference time.

$$T_{CLK} = 1/f_{CLK} \quad (4)$$

In order to compensate for phase shifts between the clock oscillators of different bus controllers, any bus controller must be resynchronized at the edge of the currently transmitted signal of interest. The synchronization hop width defines the maximum number of clock cycles that can be shortened or lengthened by resynchronization per bit period.

$$t_{SJW} = T_{SCL} \times (2 \times SJW.1 + 1 \times SJW.0 + 1) \quad (5)$$

Bus Timing Register 1 Defines the location of each bit cycle length sample point and the number of samples at each sample point. The bit function descriptions are listed in Table 2, where the SAM meaning is shown in Table 3.

Table 2: Bus Timing Register 0 (BTR0) Bit Function Description

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SAM	TSEG2.2	TSEG2.1	TSEG2.0	TSEG1.3	TSEG1.2	TSEG1.1	TSEG1.0

Table 3: Description of the Meaning of SAM

Position	Value	Function
SAM	0	Triple: bus sampling three times: recommended for medium/low speed buses (class A and B), with filtering of burrs on the buses.
	1	Single-fold: bus sampled once; recommended for use on high-speed buses (SAEC class)

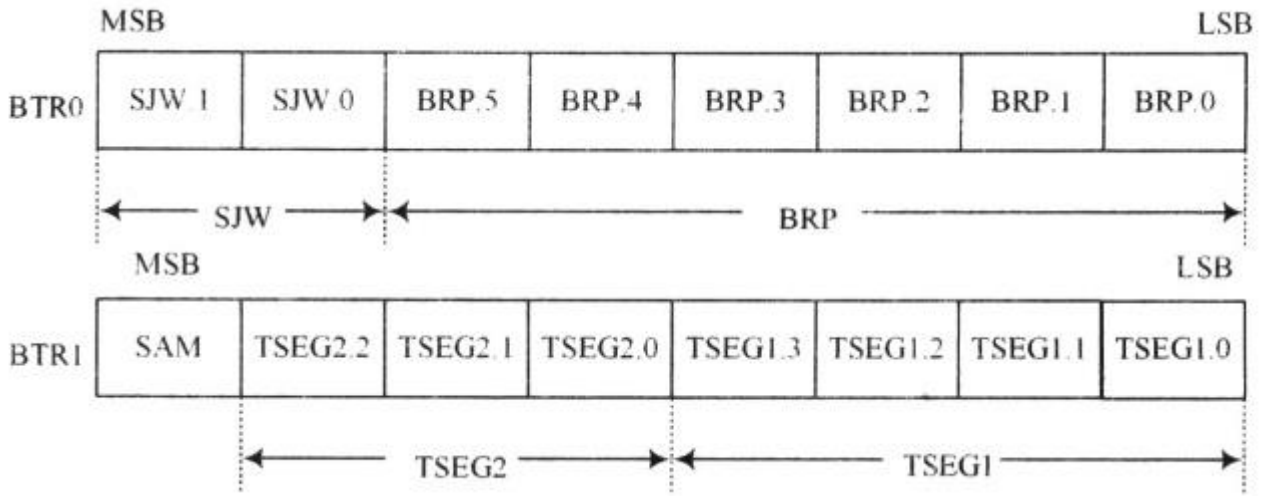


Figure 1 Bus Timing Register

T_{SEG1} (4 bits in total, value interval [1,16]) and T_{SEG2} (3 bits in total, value interval [1,8]) determine the number of clocks per bit and the location of the sampling point.

$$t_{SYNC_SEG} = 1 \times T_{SCL} \text{ (Fixed Time)} \quad (6)$$

$$t_{TSEG1} = T_{SCL} \times (8 \times TSEG1.3 + 4 \times TSEG1.2 + 2 \times TSEG1.1 + 1 \times TSEG1.0 + 1) \quad (7)$$

$$t_{TSEG2} = T_{SCL} \times (4 \times TSEG2.2 + 2 \times TSEG2.1 + 1 \times TSEG2.0 + 1) \quad (8)$$

The scalar value of the bit period (NBT) is defined as the sum of, SYNC_SEG (the number of system clock periods for the synchronization segment), TSEG1 (the number of system clock periods for phase buffer segment 1), and TSEG2 (the number of system clock periods for phase buffer segment 2). This determines its value interval to be [3,25], with the minimum value generally taken as 4 at 1 sampling point and 5 at 3 sampling points.

$$NBT = T_{bit} / T_{SCL} = SYNC_SEG + T_{SEG1} + T_{SEG2} \quad (9)$$

The general structure of the bit cycle is shown in figure 2

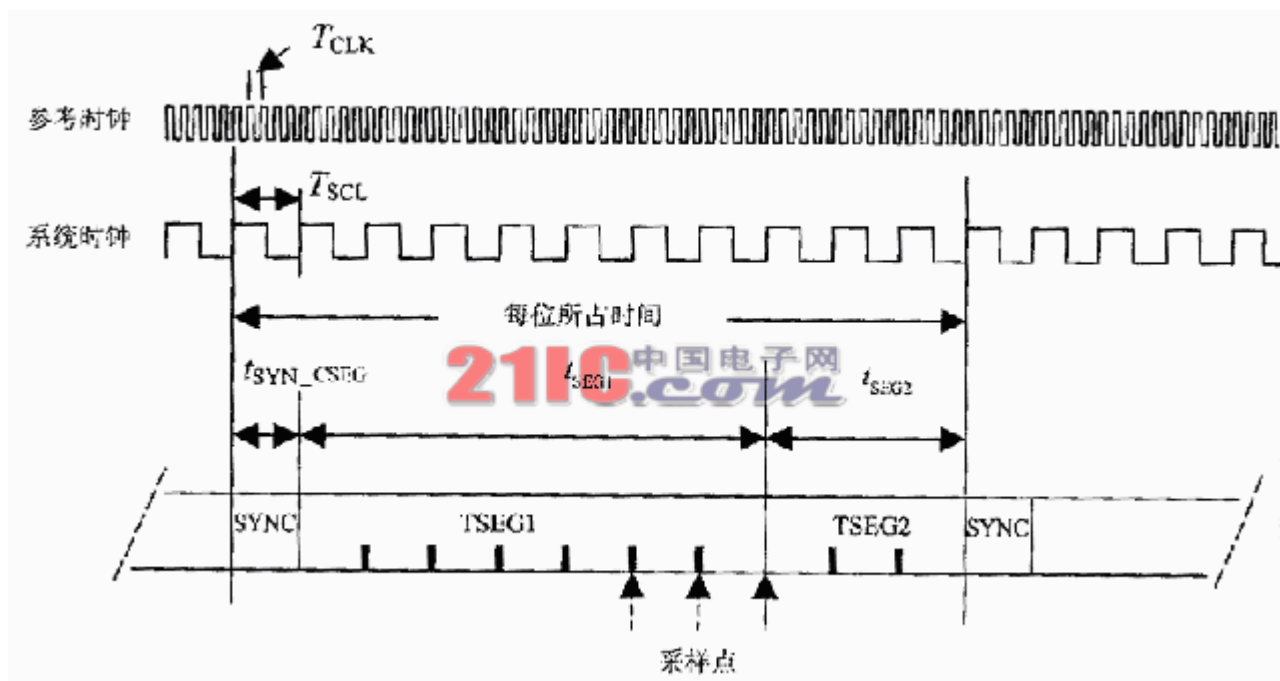


Fig. 2 Bit-cycle structure diagram

3. Calculation of CAN Baud Rate

Actually, the baud rate calculation of CAN is very simple, just that we unconsciously complicate the simple problem.

Suppose we first disregard the SJW bit in BTR0 and the SAM bit in BTR1. Then, BTR0 and BTR1 are 2 crossover coefficient registers; their product is an extended crossover coefficient. To wit:

$$BTR0 \times BTR1 = F_BASE / F_{bps} \quad (10)$$

Among them:

Internal frequency reference source $F_BASE = F_{clk}/2$, which is the 2-division frequency of the external crystal frequency F_{clk} . Note that any application, when utilizing an external crystal as a reference source, is first shaped by a 2-division frequency.

In the formula (10), when the crystal is 16M, $F_BASE = 8000K$, when the crystal is 12M, $F_BASE = 6000K$, F_{bps} is the CAN bus frequency we want. The unit is K.

Setting $BTR0 = m$ and $BTR1 = n$ in equation (10), with an external crystal of 16M,

$$m \times n = 8000 / F_{bps} \quad (11)$$

In this way, when F_{bps} takes the value we want, we get a combination of $m \times n$ values. When n is selected, the m value is also unique.

The CAN specification for the value of n provides for 8 to 25. (That is, the value of $BTR1$.) The basic principle is that the higher the value of F_{bps} , the larger the value of n (by setting $BTR1$) is selected. The reason for this is not difficult to understand.

I assume that $n = 10$ is chosen for general applications, that is:

$$\begin{aligned} &\text{Synchronization segment} + \text{phase buffer segment 1} + \text{phase buffer} \\ &\text{segment 2} = 1 + 5 + 4 \end{aligned}$$

Then equation (11) simplifies to

$$m = 800 / F_{bps}$$

The maximum setting value of m is 64, and the maximum crossover coefficient of SJA1000 $m \times n = 64 \times 25 = 1600$. Therefore, the standard algorithm usually takes 16M crystal as an example. In fact, with formula (10), any crystal value (6M to 24M) is easy to calculate.

Determination of SAM: at low frequencies, pick $SAM = 1$, i.e., sample 3 times. At high frequencies above 100K, select $SAM = 0$, i.e., sample 1 time.

SJA resynchronization jump width selection: related to the digital phase-locked loop technology. n value is selected large, SJA can be selected large, that is, at a time can be corrected for more than one pulse share T_{scl} . n value is small or the frequency is low, select $SJA = 1$. that is, $BTR0.7$ and $BTR0.6$ are set to 0.

4. Schedule: Commonly Used Baud Rate and Bus Timing Register Settings

Comparison Table

CAN Baud	Timing0(BTR0)	Timing1(BTR1)
10 Kbps	0x31	0x1C
20 Kbps	0x18	0x1C
40 Kbps	0x87	0xFF
50 Kbps	0x09	0x1C
80 Kbps	0x83	0xFF
100 Kbps	0x04	0x1C
125 Kbps	0x03	0x1C
200 Kbps	0x81	0xFA
250 Kbps	0x01	0x1C
400 Kbps	0x80	0xFA
500 Kbps	0x00	0x1C
666 Kbps	0x80	0xB6
800 Kbps	0x00	0x16
1000 Kbps	0x00	0x14
33.33 Kbps	0x09	0x6F
66.66 Kbps	0x04	0x6F
83.33 Kbps	0x03	0x6F